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Can You Stop Elly De La Cruz from Stealing Second Base?

This notebook investigates the relationship between stolen base metrics and stolen base outcomes for Elly De La Cruz, with the goal of identifying strategies to prevent him from stealing second base. It contains the code used to generate plots and preliminary findings. A more detailed explanation of methodology and results is provided in the accompanying PDF report.

1. Imports

We begin by importing the datasets and Python packages required for analysis. Data tables were sourced from **Baseball Savant**, which provide baserunning, catching, and pitching statistics.

In []:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In []:

```
'''      Elly de la cruz data      '''
elly = pd.read_csv("./ellydelacruz.csv")
elly = elly.drop(columns=['Runner', 'Fielder', 'pitcher_stealing_runs', 'Ball', 'Strike', 'Link'])
elly = elly.sort_values(by = "at_pitch_release")
elly = elly.sort_values(by = "SB/CS")
elly["Jump"] = np.nan
elly['Jump'] = elly['Jump'].astype(float)
```


2. Data Manipulation

2.1 Creating New Statistics

To obtain the *Jump* metric, we subtracted *at_pitch_release* from *lead_distance_gained* for each row. A new column, *Jump*, was added to the dataframe.

```
'''      Fill Jump      '''
for i, row in elly.iterrows():
    elly.loc[i, 'Jump'] = elly.loc[i, 'at_pitch_release'] - elly.loc[i, 'lead_distance_gained']
```

2.2 Creating Dataframes

We split the dataset into three subsets:

- Successful attempts: De La Cruz successfully stole a base
- Caught stealing: De La Cruz was thrown out
- Pitchouts: The defense attempted a pitchout

```
'''      Print table with pitchout occurrences when Elly steals      '''
elly_po = elly.copy()
elly_po.set_index('Notes')
for i, row in elly_po.iterrows():
    temp = row['Notes']
    base = row['SB/CS']
    if temp != 'Pitchout' or (temp == 'Pitchout' and base == 'SB'):
        elly_po.drop(i, axis = 0, inplace = True)
```

```
'''      Print table with only Elly's Stolen Base occurrences      '''
elly_sb = elly.copy()
elly_sb.set_index('SB/CS')
for i, row in elly_sb.iterrows():
    temp = row['SB/CS']
    if temp != 'SB' :
        elly_sb.drop(i, axis = 0, inplace = True)
```

```
'''      Print table with only Elly's caught stealing occurrences      '''
elly_cs = elly.copy()
elly_cs.set_index('SB/CS')
for i, row in elly_cs.iterrows():
    temp = row['SB/CS']
    if temp == 'SB':
        elly_cs.drop(i, axis = 0, inplace = True)
```

3. Results and Plots

The goal of this project was to evaluate whether stolen base success could be predicted or contained. Using the engineered statistics and subsets, we generated several plots:

- **Pitch velocity vs. lead distance vs. stolen base outcome**

This block of code creates a plot that gets the relationship between the velocity of the pitch, the lead distance from 1B at pitch release, and the stolen base outcome.

```
'''      Velocity of ball VS Distance from 1B at Pitch Release      '''
ax = elly_sb.plot(kind = "scatter", x='Velocity', y='at_pitch_release', label='Stolen Base', color='green', title = "Velocity of Pitch vs Distance from 1B at Pitch Release")
elly_cs.plot(kind = "scatter", x='Velocity', y='at_pitch_release', label='Caught Stealing', color='red', ax=ax)
elly_po.plot(kind = "scatter", x='Velocity', y='at_pitch_release', label='Pitchout', color='blue', ax=ax)
ax.set_xlabel("Velocity of Pitch (mph)")
ax.set_ylabel("Distance from 1B at Pitch Release (ft)")
plt.show()
```

- **Pitcher handedness vs. lead distance vs. outcome**

This block of code creates a plot that gets the relationship between the Handedness of the pitcher, the lead distance from 1B, and the stolen base outcome.

```
'''      Handedness vs Lead Distance      '''
ax = elly_sb.plot(kind = "scatter", x='Handedness', y='lead_distance_gained', label='Stolen Base', color='green', title = "Handedness vs Lead Distance")
elly_cs.plot(kind = "scatter", x='Handedness', y='lead_distance_gained', label='Caught Stealing', color='red', ax=ax)
elly_po.plot(kind = "scatter", x='Handedness', y='lead_distance_gained', label='Pitchout', color='blue', ax=ax)
ax.set_xlabel("Handedness")
ax.set_ylabel("Lead Distance from 1B (ft)")
plt.show()
```

This block of code creates a plot that gets the relationship between the Handedness of the pitcher, the distance from 1B at pitch release, and the stolen base outcome.

- **Pitcher handedness vs. distance at pitch release vs. outcome**

```
'''      Handedness vs Distance from 1B at pitch release      '''
ax = elly_sb.plot(kind = "scatter", x='Handedness', y='at_pitch_release', label='Stolen Base', color='green', title = "Handedness vs Distance from 1B at Pitch Release")
elly_cs.plot(kind = "scatter", x='Handedness', y='at_pitch_release', label='Caught Stealing', color='red', ax=ax)
elly_po.plot(kind = "scatter", x='Handedness', y='at_pitch_release', label='Pitchout', color='blue', ax=ax)
ax.set_xlabel("Handedness")
ax.set_ylabel("Distance from 1B at Pitch Release (ft)")
plt.show()
```

- **Runner's jump vs. outcome**

This block of code creates a plot that gets the relationship between the jump that Elly De La Cruz gets and the stolen base outcome.

```
'''      Jump vs Stolen Base Outcome      '''
ax = elly_sb.plot(kind = "scatter", x='Jump', y='SB/CS', label='Stolen Base', color='green', title = "Jump vs Stolen Base Outcome")
elly_cs.plot(kind = "scatter", x='Jump', y='SB/CS', label='Caught Stealing', color='red', ax=ax)
elly_po.plot(kind = "scatter", x='Jump', y='SB/CS', label='Pitchout', color='blue', ax=ax)
ax.set_ylabel("Outcome")
plt.gca().invert_yaxis()
ax.set_xlabel("Jump (ft)")
plt.show()
```

5. Acknowledgements

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